Linear Potentiometer Filter Design

Jonathan Rusnak

August 2024

Contents

1 Introduction **1**
1.1 The Purpose . . . 1
1.2 Requirements . . . 1
1.3 Measurement Resolution . . . 2

2 Design Theory **2**
2.1 Overview . . . 2
2.2 Selecting the Corner Frequency of the Digital Filter . . . 2
2.3 Digital Filtering Theory . . . 3
2.3.1 A Note about Convolution . . . 3
2.3.2 Selecting the transfer function . . . 4
2.3.3 Applying a Window Function . . . 5
2.4 Digital Filter Design . . . 6
2.5 Anti-Aliasing Filter . . . 8
2.5.1 16 Bit Design at with 10 kHz Nyquist Frequency . . . 8
2.5.2 Time Domain Distortion and the Band of Interest . . . 8
2.5.3 Test Results of the Physical Anti-Aliasing Filter . . . 9
2.6 Answers to Closing Questions on Earlier Versions of this Document . . . 11
2.6.1 Answer to Question 1 . . . 11
2.6.2 Answer to Question 2 . . . 11
2.6.3 Answer to Question 3 . . . 11
2.7 Closing Questions . . . 12
2.8 Future Work . . . 12
2.8.1 Sigma Delta Conversion . . . 12

1 Introduction

For this paper, I will design a multi stage filter with an analog and digital stage.

1.1 The Purpose

The filter is being used to process the voltage signal from the linear potentiometers on the Formula SAE internal combustion car at the University of Texas at Dallas.

1.2 Requirements

The filter is subject the following requirements:

- The group delay must be flat for the entire spectrum.
- The corner frequency of the digital filter will be 50 Hz.
- The damped resonant frequency of the car has been estimated at around 25 Hz. Note that the undamped natural frequency is less than 5 Hz.

- The DSP must be done in real time on the SUS Hat. This was originally to be the WBSS, but the WBSS design will take a two year design cycle. The implementation of the WBSS will go on the 2026 car.
- The SUS Hat will use an external 16 bit Successive Approximation Register (SAR) ADC.
- The signal must be sampled as fast as possible to lower the quantization noise floor and to ease the requirements of the anti-aliasing filter.
- Unity gain
- **Anti-Aliasing Filter Requirements**
  - The magnitude attenuation must be $20\log(2^{-n})$ dB at the Nyquist frequency.
  - The filter must leave the signal un-perturbed in the passband of the digital filter: No phase shift, and no magnitude attenuation.
  - The filter's impedance must be as high as possible.
  - Avoid electrolytic capacitors, so the capacitors values must be small values, preferably in the nF range.
  - Use a few components as possible.
  - Passive components will be used to limit the deformation of the signal in the passband of the digital filter.
  - The traces must be far apart and as short as possible. The equation shows us that we need to limit the exposed area between the traces, to prevent capacitive coupling between the traces:

$$C = \frac{\epsilon_0 A}{d}$$

1.3 Measurement Resolution

With 3.6V to work with, a 16 bit number would give a voltage resolution of:

$$\frac{3.6}{2^{16}} \approx 54.9\ \mu\text{V}$$

The spatial resolution is:

$$\frac{l}{R} = \frac{0.075}{2^{16}} \approx 1.44\ \mu\text{m}$$

2 Design Theory

2.1 Overview

The filter will have two stages: an anti-aliasing filter, and a digital filter. The anti-aliasing filter's job is to make any signal at or above the Nyquist frequency impossible for the ADC to resolve, which will prevent high frequency noise from aliasing into the signal. The anti-aliasing filter will be a low pass analog filter with a corner frequency well above the band of interest, that way the phase of the signal is not perturbed when it passes through the analog stage of the filter. The digital filter will do the heavy lifting with a brick-wall amplitude response and a linear phase response.

2.2 Selecting the Corner Frequency of the Digital Filter

We calculated the corner frequency of the filter based on the maximum speed that the linear potentiometer is capable of traveling at. Per the data sheet, the maximum velocity of the plunger is:

$$v_{\text{max}} = 10\frac{\text{m}}{\text{s}}$$

By dividing by the length of the potentiometer, we are left with a mathematical statement of the minimum corner frequency that the filter can be set at:

$$f_{\text{min}} = \frac{v_{\text{max}}}{l} \approx 133.33\ \text{Hz}$$

Though this is the minimum frequency for which the lin-pots could theoretically output given their maximum velocity and length, we are not interested in frequencies quite this high. We need to characterize the plane of the car in space and time, rather than the suspension's performance over every bump in the road. In addition, the sampling rate of the filtered signal is much lower (100 Hz), which implies that the corner frequency of the digital filter should be:

$$\boxed{\frac{100\ \text{Hz}}{2} = 50\ \text{Hz}}$$

2.3 Digital Filtering Theory

The digital filter will follow the following mathematical algorithm:

$$\overrightarrow{v}_{\text{out}}(t) = \overrightarrow{h}(t) * \overrightarrow{v}_{\text{in}}(t)$$

The equation is the convolution of the input signal vector with a vector that describes the impulse response of the transfer function of the digital filter.

This equation is derived from the mathematical principle that convolution in the time domain is multiplication in the frequency domain. First, we begin with a typical definition of a transfer function:

$$\frac{\mathbf{V}_{\text{out}}(s)}{\mathbf{V}_{\text{in}}(s)} = \mathbf{H}(s)$$

Rearranging the function algebraically gives:

$$\frac{\mathbf{V}_{\text{out}}(s)}{\mathbf{V}_{\text{in}}(s)} = \mathbf{H}(s) \rightarrow \mathbf{V}_{\text{out}}(s) = \mathbf{H}(s)\mathbf{V}_{\text{in}}(s)$$

Taking the inverse Laplace transform to convert from the time domain to the frequency domain:

$$\mathcal{L}^{-1}\left\{\mathbf{V}_{\text{out}}(s) = \mathbf{H}(s)\mathbf{V}_{\text{in}}(s)\right\} \rightarrow$$

$$\overrightarrow{v}_{\text{out}}(t) = \overrightarrow{h}(t) * \overrightarrow{v}_{\text{in}}(t)$$

We know that the transfer function is the impulse response of the filter because:

$$(1)\frac{\mathbf{V}_{\text{out}}(s)}{\mathbf{V}_{\text{in}}(s)} = (1)\mathbf{H}(s)$$

$$\mathcal{L}^{-1}\left\{(1)\mathbf{H}(s)\right\} = h(t) * \delta(t) = h(t)$$

Now that we have considered the continuos case, we will extend it to the discrete case. The discrete version of the Laplace transform is the z-transform. Given the filter equation:

$$\mathbf{V}_{\text{out}}(s) = \mathbf{H}(s)\mathbf{V}_{\text{in}}(s)$$

$$\mathcal{Z}^{-1}[\mathbf{V}_{\text{out}}(s) = \mathbf{H}(s)\mathbf{V}_{\text{in}}(s)]$$

$$\boxed{v_{\text{out}}(n) = h(n) * v_{\text{in}}(n)}$$

2.3.1 A Note about Convolution

Re-writing it with discrete values in mind:

$$(a * b)_n = \sum_{i,j} a_i \cdot b_j$$

Where $i + j = n$ for all i and j.

$$v_{\text{out}_n} = \sum_{i,j} h_i \cdot v_{\text{in}_j}$$

Notice that each index n is a dot product, where the indices align between each vector. Suppose the following vectors:

$$\overrightarrow{a} = [a_1, a_2, a_3]$$
$$\overrightarrow{b} = [b_1, b_2, b_3]$$

$$\begin{aligned}
v_{\text{out}_1} &= a_1 \cdot b_1 \\
v_{\text{out}_2} &= a_1 \cdot b_2 + a_2 \cdot b_1 \\
v_{\text{out}_3} &= a_1 \cdot b_3 + a_2 \cdot b_2 + a_3 \cdot b_1 \\
v_{\text{out}_4} &= a_2 \cdot b_3 + a_3 \cdot b_2 \\
v_{\text{out}_5} &= a_3 \cdot b_3
\end{aligned}$$

$$\overrightarrow{v}_{\text{out}} = [a_1 \cdot b_1, a_1 \cdot b_2 + a_2 \cdot b_1, a_1 \cdot b_3 + a_2 \cdot b_2 + a_3 \cdot b_1, a_2 \cdot b_3 + a_3 \cdot b_2, a_3 \cdot b_3]$$

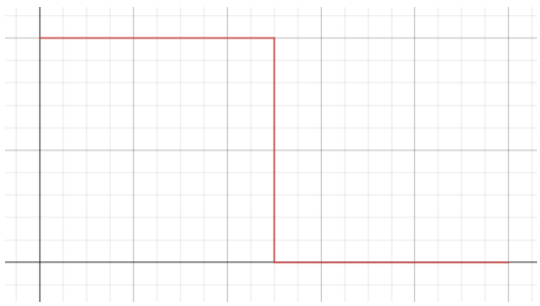
Written as a column vector:

$$\overrightarrow{v}_{\text{out}} = \begin{bmatrix} a_1 \cdot b_1 \\ a_1 \cdot b_2 + a_2 \cdot b_1 \\ a_1 \cdot b_3 + a_2 \cdot b_2 + a_3 \cdot b_1 \\ a_2 \cdot b_3 + a_3 \cdot b_2 \\ a_3 \cdot b_3 \end{bmatrix}$$

Since the phase response of the digital filter is linear, we will know the broad spectrum delay, and this allows us to shift the index of the processed information to get rid of the delay in our signal.

2.3.2 Selecting the transfer function

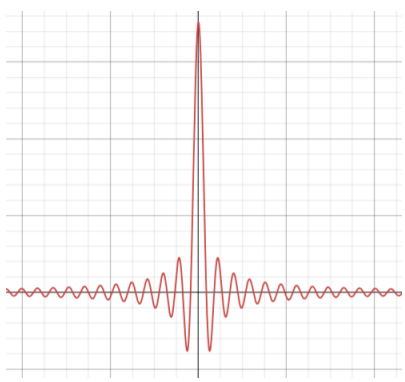
In the frequency domain, the ideal low pass filter is characterized by the following magnitude response:



Where the horizontal axis is frequency, and the vertical axis is the amplitude.

Applying the z-transform gives a sync function:

$$\mathcal{Z}^{-1}\left\{\mathbf{H}(s)\right\} = h(n) = \frac{\sin(n)}{n}$$



This implies the relationship:

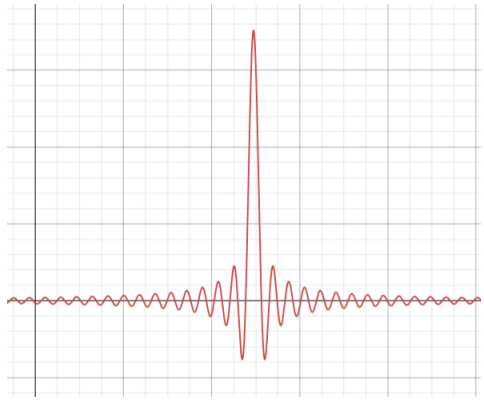
$$v_{\text{out}}(n) = h(n) * v_{\text{in}}(n) \rightarrow$$

$$v_{\text{out}}(n) = \frac{\sin(n)}{n} * v_{\text{in}}(n)$$

To prevent the use of negative indices and induce a more convenient symmetry, we shift the function by the filter order M over 2:

$$v_{\text{out}}(n) = \frac{\sin\left[n - \frac{M}{2}\right]}{n - \frac{M}{2}} * v_{\text{in}}(n)$$

Which gives:



This fact also allows us to easily create a filter with a linear phase response. $h(n)$ must be chosen such that the numbers are symmetrical about $M/2$. This symmetry yields a broad spectrum linear phase response.

To modulate the desired corner frequency, we multiply the argument to the sin function with the corner frequency ω_c:

$$v_{\text{out}}(n) = \frac{\sin\left[\omega_c\left[n - \frac{M}{2}\right]\right]}{n - \frac{M}{2}} * v_{\text{in}}(n)$$

$$\omega_c = 2\pi f_c$$

$$v_{\text{out}}(n) = \frac{\sin\left[2\pi f_c\left[n - \frac{M}{2}\right]\right]}{n - \frac{M}{2}} * v_{\text{in}}(n)$$

Note that the corner frequency f_c is normalized to the sampling frequency f_s. The value of f_c is the ratio of the desired half amplitude frequency to the sampling frequency:

$$f_c = \frac{f_h}{f_s}$$

This ratio is a critical design consideration for digital filters.

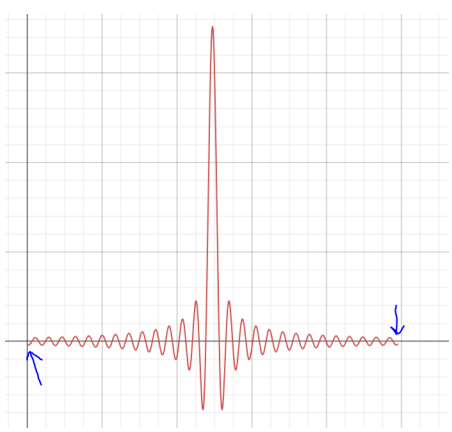
Since the corner frequency of the digital filter is so low, this heavily bumps up the requirements of the anti-aliasing filter. In fact, since the nyquist frequency of the ADC is 10 kHz and the resolution is 16 bits, we need an attenuation of:

$$A = 20\log(2^{-16}) \approx -96.3 \text{ dB}$$

This is an extremely tall order for any analog filter because of the combination of passband and stop band requirements.

2.3.3 Applying a Window Function

Wouldn't it be nice if it was that easy? Mathematically speaking, this is a perfectly legitimate approach. Unfortunately, this equation must be computed by a computer, and not a human. Therefore, we cannot consider infinite points, and we must truncate the signal.



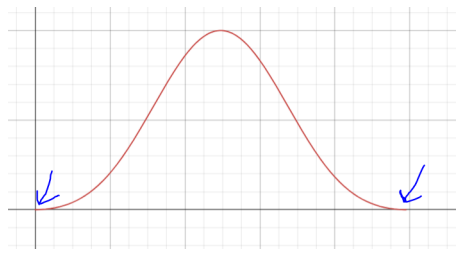
The blue arrows indicate an abrupt end to the sync function. This presents yet another problem! The truncation causes instability (ripples) in the frequency domain. This translates to an injection of broad-spectrum noise caused by the nature of our processing method!

As a result, we must apply a window function to smooth out the truncations. In this filter, we apply a Blackman window function:

$$w(n) = 0.42 - 0.5\cos\left[\frac{2\pi}{M}n\right] + 0.08\cos\left[\frac{4\pi}{M}n\right]$$

Where M is the order of the filter.

Notice that the Blackman function approaches zero at the limits of the filter:



This implies that using the Blackman function as an amplitude modulator will give the desired sync function. Recall the original equation:

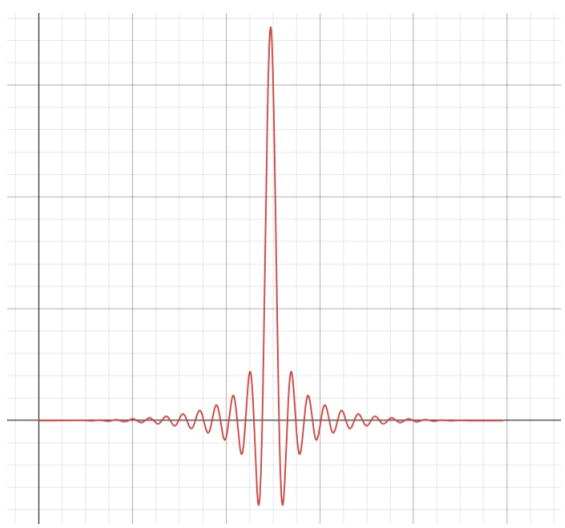
$$v_{\text{out}}(n) = h(n) * v_{\text{in}}(n)$$

$$v_{\text{out}}(n) = h(n)w(n) * v_{\text{in}}(n)$$

Note that we do not multiply both sides by the window function, since the point is to actually change the frequency response. This gives the final function for the digital filter to evaluate:

$$\boxed{v_{\text{out}}(n) = \left[K\frac{\sin\left[2\pi f_c\left(n - \frac{M}{2}\right)\right]}{n - \frac{M}{2}}\right]\left[0.42 - 0.5\cos\left[\frac{2\pi}{M}n\right] + 0.08\cos\left[\frac{4\pi}{M}n\right]\right] * v_{\text{in}}(n)}$$

The resulting sync function:



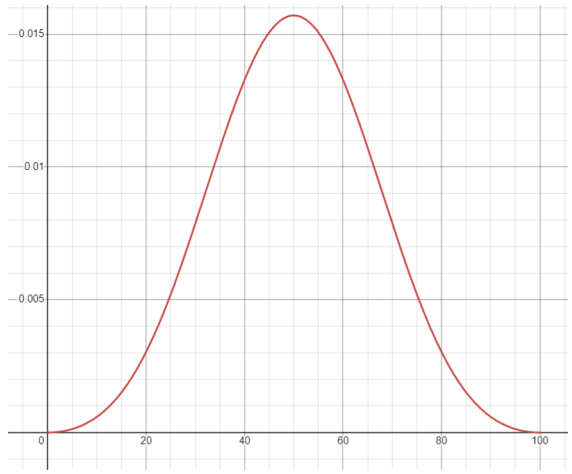
2.4 Digital Filter Design

The main design parameters are as follows:

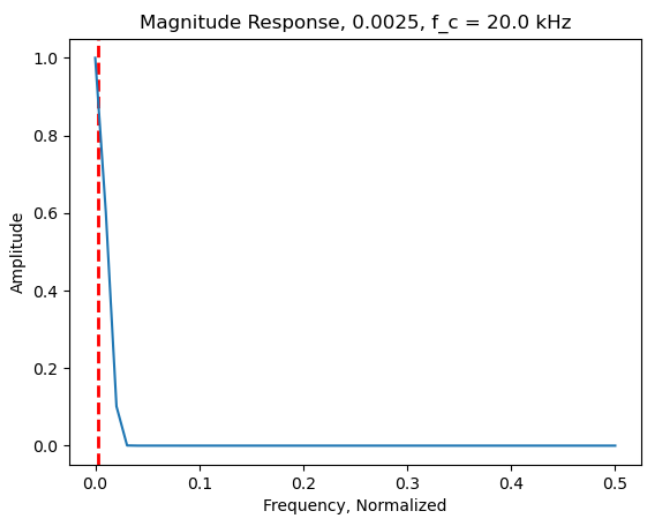
- The half amplitude frequency is 50 Hz.
- The order is 100.
- The sampling rate is 20 kHz.

$$\boxed{v_{\text{out}}(n) = \frac{1}{n-50}\sin\left[0.0157\,(n-50)\right]\left[0.42 - 0.5\cos\left[0.02\pi n\right] + 0.08\cos\left[0.04\pi n\right]\right] * v_{\text{in}}(n)}$$

The graph of the sync function is shown below, where the y axis is the amplitude and the x axis is samples:



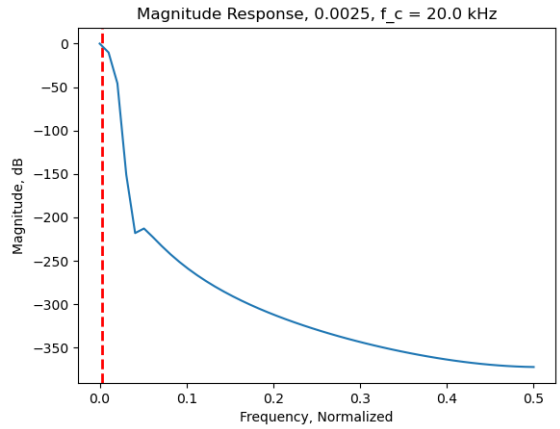
In simulation, the amplitude response with respect to frequency is:



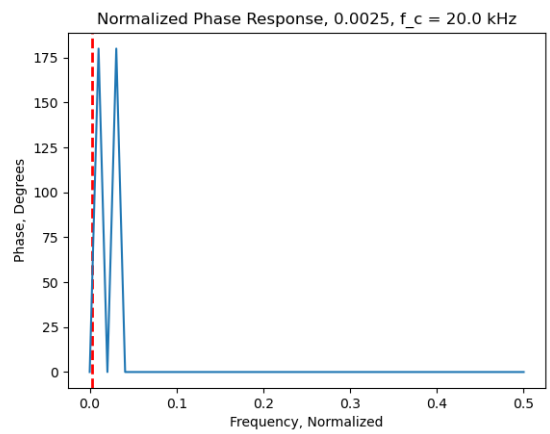
The red line indicates the location of the corner frequency. Normally, the intersection lies at:

$$(f_c, 0.5)$$

However, given the requirements, it will not be possible to do that with a singe digital filter stage. More on this later. The lowest corner is at about 600 Hz. Observing the logarithmic plot, we get a better idea of the attenuation levels:



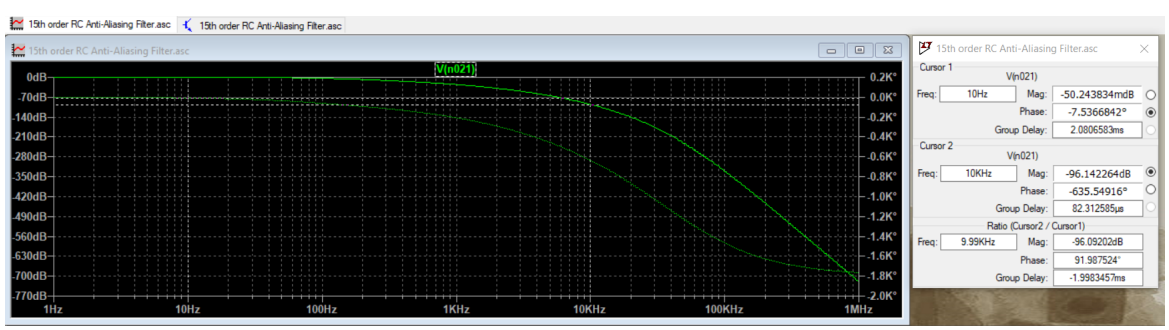
The first corner is at about 800 Hz, which translates to more that 200 dB per decade! The phase response is linear. Note that the phase is wrapped:



2.5 Anti-Aliasing Filter

2.5.1 16 Bit Design at with 10 kHz Nyquist Frequency

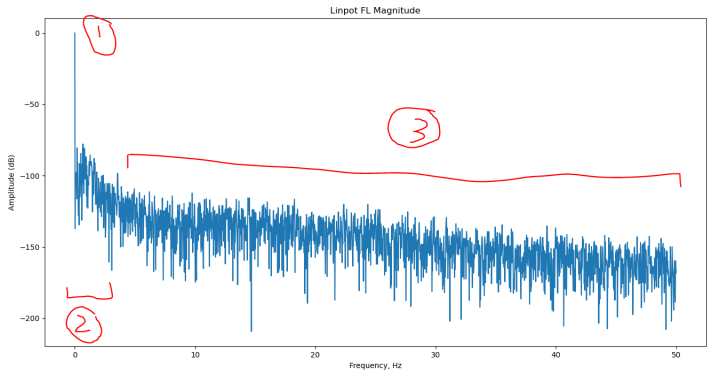
With a 10 kHz nyquist frequency, we have implemented a 20th order RC low pass filter. The filter has a ladder architecture, with resistor values at 1 kΩ and capacitor values at 1 nF. The frequency response of which is:



The magnitude response in in spec at -96 dBs. The phase at 10 Hz is at -7 degrees, which is not optimal. The size of the filter is also not optimal, since it requires 40 components. However, we were able to use 0402 package parts that are soldered on at the factory. This yielded a very small footprint for the filter, which sits under an EMI shield.

2.5.2 Time Domain Distortion and the Band of Interest

Given the constraints, I needed to better evaluate the spectrum of interest. I was able to get my hands on the data from the 2024 car, the spectrum of which is shown below:



The artifact descriptions are as follows:

1. The DC component has a large magnitude compared to the rest of the signal. This is to be expected, and the same artifact will show up in the 2025 car's data.

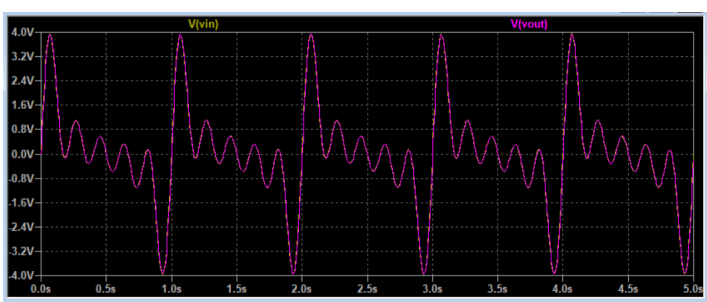
2. This information is just good enough to get a rough idea of the spectrum of interest, which appears to be less than 3-5 Hz.

   - Zheer mentioned that the undamped resonant frequency is slightly greater, which fits the general theory that suggests damped resonant frequency should be lower than the natural frequency of a system.

3. My theory about the shape of this graph has a few parts:

   - I think most of this information in region 3 is below the nose floor, since the relative magnitude is small.
   - The noise floor should be quite high, since the sampling rate is very low. In other words, the quantization noise injected by the low sampling rate contributes to the broad spectrum noise we see here. In addition, the ADC injects quantization noise since the LSB cannot represent the signal with infinite precision.
   - Another contributing factor will be the aliasing of other high frequency signals, such as bumps in the road (100s of Hz) and electromagnetic noise.

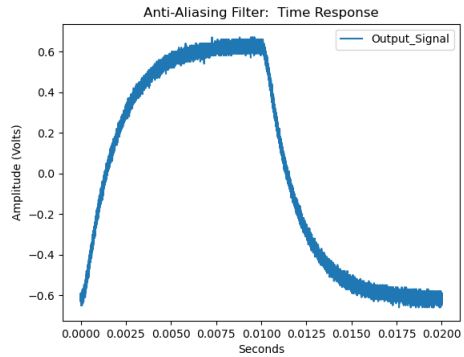
These are very important results, especially the identification of the spectrum of interest (signals less than 5 Hz). This gives us design freedom in the anti-aliasing filter, since the distortions are out of band for the 20th order filter, as shown in the LT spice simulation:



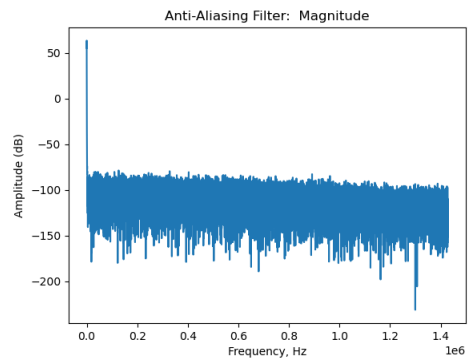
The simulation injected sinusoids at 1, 2, 3, 4, and 5 Hz. The yellow is the input, and the purple is the output signal. This shows very little time domain distortion in the spectrum of interest.

2.5.3 Test Results of the Physical Anti-Aliasing Filter

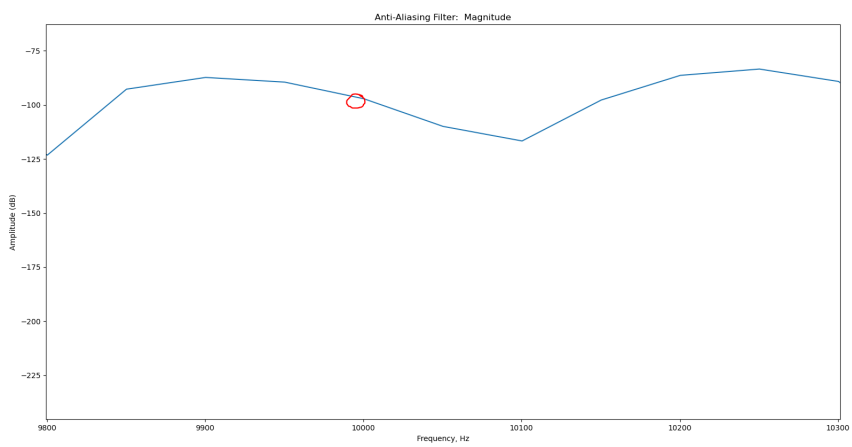
When the PCB arrived, I tested it with a 50 % duty cycle 50 Hz square wave. I captured one output waveform, which is shown below:



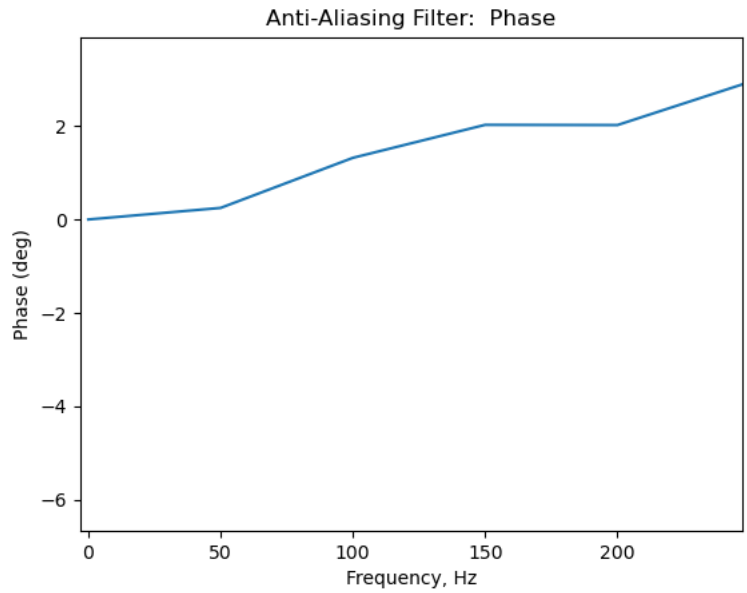
The magnitude response is:



Zooming in at 10 Hz, we see that the filter is performing within the specifications:



The surprising results come when we examine the phase response:



This shows almost no perturbation through the entire passband of the digital filter, which is highly desirable. I was expecting -7 degrees at 10 Hz based on the simulations, but the actual filter shows less than 1 degree at 50 Hz. This is highly desirable, but I don't have a strong understanding of the mechanisms for this. My theory is that the source of the positive group delay is parasitic inductance. However, this would require very large inductances since the frequencies are so low and the capacitive reactance is so high.

2.6 Answers to Closing Questions on Earlier Versions of this Document

1. The major road block is, how will we design a filter with such a low cutoff frequency requirement?
   - We can use a powerful analog filter, but the inductors are in the 100s of mH, or we would need a higher order filter.
   - We can cascade digital filters by adding an external $\Sigma\delta$ converter, using the onboard digital filter, then using the filter I designed.
2. When I do a time domain convolution with the input vector and the time domain impulse response of the transfer function, how do I correctly time stamp the resulting vector?
3. Is it more efficient to take the FFT of the input vector then do an element wise multiplication with the transfer function in the frequency domain vs a time domain convolution?

2.6.1 Answer to Question 1

We will use a powerful, high order RC filter instead. This still gives a small footprint if we use small packages. For the H7, we will cascade digital filters to reduce the size of the AA filter.

2.6.2 Answer to Question 2

Currently, I do not have an answer for this.

2.6.3 Answer to Question 3

Yes. It is more efficient to do the multiplication in the frequency domain. This has some caveats. It is much more complicated to do the multiplication in the frequency domain, since the FFTs require a specific input signal width. In chapter 18 of the DSP book (The Scientist and Engineer's Guide to Digital Signal Processing), the author notes that it is not worth using FFT convolution unless the filter kernal needs to be quite large. In this case, he suggests a filter kernal of at least 1000. I believe this will be possible using the H7.

2.7 Closing Questions

So far, the only question that remains is:

1. When I do a time domain convolution with the input vector and the time domain impulse response of the transfer function, how do I correctly time stamp the resulting vector?

2.8 Future Work

For the 2026 car, we will use nested digital filters. This will allow us to use a much smaller anti-aliasing filter, and the filter kernal will have a much better magnitude response.

2.8.1 Sigma Delta Conversion

If we use exactly the same chip for the 2026 car, we can use an external sigma delta modulator. We will feed the sigma delta modulated signal into the sigma delta interface, which has an onboard digital filter. The corner frequency of the digital filter can be set to 25 kHz, since this is 2 orders of magnitude above the nyquist frequency of the next filter, whose sampling rate is 5 kHz.

The second filter will have the same specifications as our filter, but the sync function will be a much higher quality since the sampling rate is lower.